MTH1002

UNIVERSITY OF EXETER

COLLEGE OF ENGINEERING, MATHEMATICS AND PHYSICAL SCIENCES

MATHEMATICS

May 2018

Mathematical Methods

Module Leader: Dr P. Philipp and Professor K. Zhang

Duration: 2 HOURS.

The mark for this module is calculated from 70% of the percentage mark for this paper plus 30% of the percentage mark for the January 2018 paper.

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.

This is a **CLOSED NOTE** examination.

SECTION A

1. (a) (i) Find the limit

$$\lim_{n\to\infty} \frac{1+3n^2}{7n^2+5}.$$

(ii) Find the radius of convergence of the power series

$$\sum_{k=0}^{\infty} \frac{x^k}{3^{k+1}}.$$

(8)

(b) Find all eigenvalues and eigenvectors of the matrix

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}.$$

(8)

(c) Find the following two indefinite integrals

$$\text{(i)} \int \left(\cos\frac{x}{2} - \sin\frac{x}{2}\right)^2 dx, \qquad \text{(ii)} \int \frac{dx}{\sqrt{x^2-4}}.$$

Hint: For part (ii), use the substitution $x = 2\cosh t$. (10)

(d) Express the definite integral

$$\int_0^2 (2x+1)\,dx$$

as a right-hand Riemann sum and then evaluate it. (8)

(e) Solve the ordinary differential equation

$$\frac{d^2y}{dx^2} + y = e^{-x},$$

subject to the initial conditions $y(0) = 1$ and $y'(0) = 0$. (8)

(f) Transform the partial differential equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial x \partial y} - 2\frac{\partial^2 u}{\partial y^2} = 0$$

to the form

$$\frac{\partial^2 u}{\partial \xi \partial \eta} = 0,$$

where $\xi = x + y$ and $\eta = x - \frac{y}{2}$. Hence, find the general solution of the equation. (8)

[**50**]

SECTION B

2. (a) Suppose that the matrix $A = (a_{ij})_{2\times 2}$ is given by

$$A = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}.$$

First, compute A^2 and A^3. Second, guess an expression for A^n with $n > 3$ and, then, prove it. (10)

(b) Consider a system of the linear equations

$$\begin{bmatrix} \alpha & 1 & 1 \\ 1 & \alpha & 1 \\ 1 & 1 & \alpha \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \alpha - 3 \\ -2 \\ -2 \end{bmatrix}.$$

Determine the values of α such that the system has (i) no solution, (ii) a unique solution and (iii) an infinite number of solutions. Hint: Change the augmented matrix to Row Echelon Form to reveal its rank as a function of α. (15)

[**25**]

3. (a) Find $f(x)$ that satisfies the first-order differential equation

$$f'(\sin^2 x) = \cos^2 x + \tan^2 x, \quad 0 < x < 1.$$

Hint: Note that $\tan^2 x = \sin^2 x/(1 - \sin^2 x)$. (10)

(b) Evaluate the integral

$$f(x) = \int_0^{\sin^2 x} \sin^{-1}\sqrt{t}\, dt + \int_0^{\cos^2 x} \cos^{-1}\sqrt{t}\, dt, \quad 0 < x < \frac{\pi}{2}.$$

Hint: First, show that

$$\frac{df(x)}{dx} = 0,$$

which implies that $f(x)$ is a constant, and then evaluate $f(x)$ at $x = \pi/4$. (15)

[**25**]

4. Suppose that $z = f(x, y)$ is differentiable and satisfies the partial differential equation

$$x^2 \frac{\partial^2 z}{\partial x^2} - y^2 \frac{\partial^2 z}{\partial y^2} = 0.$$

Solve this equation via the following three steps.

(a) First, introduce the new independent variables u and v, $u = xy$ and $v = x/y$, to transform the equation to

$$\frac{\partial}{\partial v}\left[\frac{\partial z}{\partial u} - \frac{z}{2u}\right] = 0.$$

(12)

(b) Second, derive a first-order partial differential equation from the equation in 4(a). (3)

(c) Third, solve the first-order partial differential equation to show that

$$z(x, y) = f(xy) + B(x/y)\sqrt{xy}$$

where

$$f(t) = \sqrt{t} \int \frac{C(t)}{\sqrt{t}}\, dt$$

and $B(t)$ and $C(t)$ are arbitrary functions of t. (10)

[**25**]